# Homework 1 Text Classification

## Bag of Words Word Embeddings and BERT

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Six text classifiers were evaluated on a shared NYT test set: two Bag of Words models, three averaged word embedding models, and a fine tuned BERT model. Word Frequency achieved the highest measured test Macro-F1 of 0.9729 with Accuracy of 98.87%. The comparison evaluates the required representations under a common data split, while preserving their prescribed training procedures.

### 1 Data and evaluation protocol

The supplied NYT file contains 11,519 labeled news articles in three categories. AG News contains 90,000 unlabeled texts and is used only to learn one Word2Vec model. All rows are retained. NYT is shuffled and stratified with random seed 42 into approximately 80% training, 10% validation and 10% test data. The same saved row indices are reused in all six experiments.

| Split | Business | Politics | Sports | Total |
|---|---|---|---|---|
| Train | 1143 | 1161 | 6911 | 9215 |
| Validation | 143 | 145 | 864 | 1152 |
| Test | 143 | 145 | 864 | 1152 |

The training set is the only source of the BoW vocabulary, NYT Word2Vec embeddings, feature standardization statistics and supervised model parameters. Validation Macro-F1 selects the Logistic Regression regularization strength and the BERT checkpoint. Test labels do not select hyperparameters, epochs or preprocessing choices. Models are not refitted on validation rows.

### Evaluation metrics

Accuracy is the fraction of correct predictions. For class k, precision is TP/(TP+FP), recall is TP/(TP+FN), and F1 is 2PR/(P+R). Macro-F1 is the unweighted mean of the three class F1 scores, with zero division scored as zero. This matters because sports accounts for 75% of the test set. An always-sports classifier reaches 75.00% Accuracy but only 0.2857 Macro-F1.

### Repeated texts

NYT contains 72 duplicate text rows. Under the prescribed row-level shuffle, 14 distinct exact texts occur in both training and validation, 10 in training and test, and 1 in validation and test. The row indices are disjoint, but the repeated content weakens the independence of evaluation. All methods face the same split, and the overlap is disclosed rather than silently removing rows.

# 2 Methods and training settings

### Two Bag of Words representations

Text is lowercased and split into Unicode word tokens using the regular expression \b\w+\b. Single-character words and numbers are retained. There is no stemming, stopword filtering or n-gram expansion. A training-only vocabulary of 61,021 words defines a sparse document vector. Binary BoW sets x_i to 1 if word i occurs and 0 otherwise. Word Frequency sets x_i to the raw occurrence count. For vocabulary [apple, banana], the text 'apple apple banana' becomes [1, 1] in binary form and [2, 1] in frequency form. Neither representation is normalized or TF-IDF weighted.

Both use L2-regularized multinomial Logistic Regression with the lbfgs solver, tolerance 1e-4 and a maximum of 2,000 iterations. C is chosen from {0.01, 0.1, 1, 10} by validation Macro-F1. A smaller C means stronger regularization. Classes are unweighted, and every candidate must converge.

### Three averaged word embedding representations

Each document receives a 100-dimensional vector by averaging the embeddings of its valid token occurrences: e(d) = sum(e(w_i))/n_valid. Repeated words contribute repeatedly. Out-of-vocabulary tokens are skipped; a document with no valid tokens receives a zero vector. Each dimension is standardized using training means and standard deviations, then classified with the same Logistic Regression search as above.

**Pretrained GloVe.** The original glove.6B.100d.txt vectors are used [1]. The external vocabulary and weights are fixed. Loading the vectors needed for validation and test tokens performs a lookup in this pretrained vocabulary, without learning from those documents.

**Word2Vec on AG News.** Skip-gram embeddings are learned from all 90,000 unlabeled AG texts [2]. The model uses dimension 100, window 5, minimum count 2, 5 negative samples, subsampling 1e-3 and 5 embedding epochs. Seed 42, one worker and a stable CRC32 hash control initialization and training variation. NYT labels are used only by the downstream classifier.

**Word2Vec on NYT.** The same embedding configuration is applied to the 9,215 NYT training texts. Validation and test articles are excluded from embedding training. This isolates the comparison between an external news corpus and the task's own training corpus.

### BERT fine tuning

The specified google-bert/bert-base-uncased checkpoint is fine tuned end to end with a three-class classification head [3, 4]. WordPiece tokenization truncates and pads each input to max_length=64, including special tokens. Training lasts exactly 3 epochs, with batch size 32, AdamW learning rate 2e-5, weight decay 0.01 except bias and LayerNorm weights, 10% linear warmup, linear learning rate decay and gradient norm clipping at 1.0. CUDA mixed precision is used. After all three epochs, the checkpoint with highest validation Macro-F1 is evaluated on the test set.

# 3 Experimental results

Table 1 reports the required metrics on the same 1,152 test articles. Validation metrics are shown to document model selection. Scores are fractions except where Accuracy is shown as a percentage.

| Representation | Val Acc | Val F1 | Test Acc | Test F1 |
|---|---|---|---|---|
| Binary Bag of Words | 98.26% | 0.9595 | 98.78% | 0.9704 |
| Word Frequency | 98.35% | 0.9626 | 98.87% | 0.9729 |
| Pretrained GloVe | 98.00% | 0.9549 | 98.52% | 0.9646 |
| Word2Vec on AG News | 97.48% | 0.9418 | 98.18% | 0.9544 |
| Word2Vec on NYT | 97.92% | 0.9511 | 98.87% | 0.9718 |
| Fine tuned BERT | 97.57% | 0.9490 | 98.18% | 0.9612 |

Table 1 Accuracy and Macro-F1 for all required experiments

### Regularization selected on validation

The selected C values are Binary Bag of Words: 0.1; Word Frequency: 1; Pretrained GloVe: 0.01; Word2Vec on AG News: 0.01; Word2Vec on NYT: 0.1. Complete candidate scores and solver iteration counts are stored in the corresponding result JSON files.

### Embedding token coverage

| Representation | Train coverage | Val coverage | Test coverage | Empty test docs |
|---|---|---|---|---|
| Pretrained GloVe | 99.78% | 99.77% | 99.79% | 0 |
| Word2Vec on AG News | 97.34% | 97.26% | 97.33% | 0 |
| Word2Vec on NYT | 99.71% | 99.23% | 99.29% | 0 |

Coverage counts occurrences, including repeated words. Empty documents receive zero vectors.

### BERT validation after each epoch

| Epoch | Train loss | Val Accuracy | Val Macro-F1 |
|---|---|---|---|
| 1 | 0.2871 | 96.35% | 0.9182 |
| 2 | 0.0535 | 97.57% | 0.9490 |
| 3 | 0.0209 | 97.40% | 0.9444 |

All three epochs were completed. Epoch 2 was selected by validation Macro-F1. Test results use this selected checkpoint. No epoch was chosen using test labels.

Elapsed times exclude environment installation and resource downloads. They are measured for this run and are not a controlled speed benchmark: Binary Bag of Words 5.9s; Word Frequency 16.0s; Pretrained GloVe 7.1s; Word2Vec on AG News 47.3s; Word2Vec on NYT 84.8s; Fine tuned BERT 130.8s.

# 4 Analysis and limitations

## Representation comparison

Word Frequency improves over Binary BoW by 0.09 percentage points in Accuracy and 0.0025 in Macro-F1. The frequency model makes 13 test errors, compared with 14 for binary features. This small difference is consistent with useful repeated topical terms, but a single split does not establish a reliable general advantage. Both retain explicit lexical evidence that can be informative for news topics.

NYT-trained Word2Vec achieves +0.0174 Macro-F1 relative to AG-trained Word2Vec. The observed result is consistent with a benefit from matching the embedding corpus to the target domain. NYT has fewer documents, but document count is not token count. These results do not isolate training-token volume, news distribution or vocabulary coverage as causal factors. Mean pooling compresses a whole article into 100 numbers and discards word order and local context; this can dilute rare category-specific terms.

BERT reaches 98.18% Accuracy and 0.9612 Macro-F1. Its contextual representation differs from the fixed averages, but its input is limited to the first 64 WordPiece tokens including special tokens, while BoW and averaged embeddings use the whole text. The comparison therefore combines representation changes with different input budgets and training procedures. The observed ranking should not be generalized to unconstrained BERT models.

## Per class test F1

| Representation | Business | Politics | Sports |
|---|---|---|---|
| Binary Bag of Words | 0.9504 | 0.9625 | 0.9983 |
| Word Frequency | 0.9556 | 0.9648 | 0.9983 |
| Pretrained GloVe | 0.9444 | 0.9517 | 0.9977 |
| Word2Vec on AG News | 0.9278 | 0.9371 | 0.9983 |
| Word2Vec on NYT | 0.9547 | 0.9619 | 0.9988 |
| Fine tuned BERT | 0.9412 | 0.9481 | 0.9942 |

## BERT test confusion matrix

| Gold / Predicted | Business | Politics | Sports |
|---|---|---|---|
| Business | 136 | 5 | 2 |
| Politics | 6 | 137 | 2 |
| Sports | 4 | 2 | 858 |

## Scope of the conclusions

The test set has only 143 business and 145 politics articles, so a few mistakes can move Macro-F1. These are single-seed results without confidence intervals or significance testing. Exact text overlap across splits may make the task easier. No claim is made that the split is independent at the story level or that the ranking transfers to new publishers, dates or categories. More reliable generalization estimates would require deduplicated or grouped splits and repeated runs.

# 5 Reproducibility and references

## Execution and saved evidence

The experiments used Python 3.12.14, Windows 11 and NVIDIA GeForce RTX 4070 Laptop GPU. Main versions were scikit-learn 1.6.1, Gensim 4.4.0, PyTorch 2.8.0 with CUDA 12.8 and Transformers 4.57.1. Seed 42 is used for the data split and training. GPU floating-point results can vary across hardware and software despite fixed seeds.

README.md documents installation, resource acquisition and execution. Run prepare_assets.py to obtain the required files, then run run_experiments.py --part all. The code saves the shared split and input fingerprint in results/split.json, aggregate results in results/summary.csv, per-method settings and metrics in JSON, and test row IDs, gold labels and predictions in CSV. verify_results.py recomputes Accuracy, Macro-F1 and confusion matrices from these predictions and checks the identical test-row ordering and the three-epoch BERT configuration.

GloVe was extracted from the 100-dimensional member of the original Stanford glove.6B.zip using HTTP byte ranges and ZIP integrity checking. BERT weights were delivered through a ModelScope mirror because the original large-file CDN was unavailable. The entire downloaded safetensors file was SHA-256 checked against the official Hugging Face file at the pinned revision; it is byte-for-byte identical. The tokenizer and configuration came from that Hugging Face revision.

BERT revision: 86b5e0934494bd15c9632b12f734a8a67f723594

BERT SHA256: 68d45e234eb4a928074dfd868cead0219ab85354cc53d20e772753c6bb9169d3

GloVe SHA256: 95dde4dfd627ab26608d33e76d1195ec059734bd29089ea52cadb08d07c64544

## Use of tools and AI assistance

The implementation uses scikit-learn for sparse vectorization, Logistic Regression and metrics; Gensim for Word2Vec; and PyTorch and Hugging Face Transformers for BERT fine tuning. AI assistance was used to implement and execute the experiments and draft the report. All reported experimental measurements were obtained by running the submitted code on the supplied data.

## References

[1] Jeffrey Pennington, Richard Socher and Christopher Manning. 2014. GloVe: Global Vectors for Word Representation. EMNLP, pages 1532-1543. https://aclanthology.org/D14-1162/

[2] Tomas Mikolov, Kai Chen, Greg Corrado and Jeffrey Dean. 2013. Efficient Estimation of Word Representations in Vector Space. arXiv:1301.3781. https://arxiv.org/abs/1301.3781

[3] Jacob Devlin, Ming-Wei Chang, Kenton Lee and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. NAACL-HLT, pages 4171-4186. https://aclanthology.org/N19-1423/

[4] Hugging Face. google-bert/bert-base-uncased model card and pinned checkpoint. https://huggingface.co/google-bert/bert-base-uncased

[5] Course Homework 1 specification and the supplied HW-1 datasets.